



24ADI204-DATA SCIENCE AND VISUALIZATION

**2025-2026
FINAL REPORT**

**DEPARTMENT OF ARTIFICIAL
INTELLIGENCE AND DATA SCIENCE**

“SPOTIFY MUSIC TRENDS “

**24ADI204 - DATA SCIENCE AND
VISUALISATION**

Submitted by

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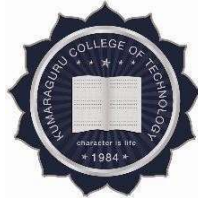
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MADHUSRI S- 24BAD065

*in partial fulfilment for
the award of the degree of*

**BACHELOR OF TECHNOLOGY
IN
ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**



KUMARAGURU COLLEGE OF TECHNOLOGY

(An Autonomous Institution affiliated to Anna University, Chennai)

Post Box No: 2034, Coimbatore - 641049

APRIL 2026

BONAFIDE CERTIFICATE

Certified that this project report “**SPOTIFY MUSIC TRENDS**” is the Bonafide work of “ADHIRAJ S - 24BAD003, JANASREE P- 24BAD040, KRITHIKA K- 24BAD061, MADHUSRI S- 24BAD065” who carried out the project work under my supervision.

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The candidates with college register number 24BAD003, 24BAD040, 24BAD061, 24BAD065 were examined in the Project Viva-Voce examination held on 16.04.2026

Internal Examiner

External Examiner

DECLARATION

I affirm that the project work titled “SPOTIFY MUSIC TRENDS” being submitted in partial fulfillment for the award of B.TECH Artificial Intelligence and Data Science is the original work carried out by us. It has not formed the part of any other project work submitted for the award of any degree or diploma, either in this or any university

ADHIRAJ S 24BAD003



JANASREE P 24BAD040



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I certify that the declaration made above by the candidates is true.

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List of abbreviations:

Abbreviation	Full Form
AI	Artificial Intelligence
DS	Data Science
EDA	Exploratory Data Analysis
PCA	Principal Component Analysis
BPM	Beats Per Minute
ML	Machine Learning
KPI	Key Performance Indicator
CSV	Comma Separated Values
API	Application Programming Interface

Executive Summary

Problem statement:

SPOTIFY MUSIC TRENDS

Dataset Link:

<https://drive.google.com/file/d/1Tg8Ps-IE0QLwRdZzDV8T1nD1aMzpvsk/view?usp=drivesdk>

Key Findings

- Most songs have medium popularity (40–60)
- Music popularity shows near normal distribution
- Songs are mostly moderately high in energy & danceability
- Tempo ranges widely (around 100–140 BPM)
- Some genres dominate → genre imbalance exists
- Music trends change over years (dynamic preferences)
- Energy & danceability slightly increase popularity
- No single feature controls popularity → multi-factor influence
- Data cleaning improved accuracy & reliability
- Feature engineering helped reduce skewness & improve model readiness

Final Insights

- Music trends show a clear preference for moderately energetic and danceable songs
- Popularity is not driven by a single factor, but by a combination of multiple audio features
- Listener preferences evolve over time, indicating dynamic music trends
- Certain genres dominate, showing uneven audience interest across genres
- Data preprocessing and feature engineering significantly improved analysis quality
- Overall, music success depends on a balance of audio characteristics and changing audience behavior.

CHAPTER 1-INTRODUCTION

Background of the problem:

The rise of digital music platforms like Spotify has made music consumption more dynamic and data driven. With thousands of songs released regularly, understanding what makes a song popular has become challenging. Listener preferences keep changing over time, influenced by trends and audio features like energy, danceability, and tempo. This project focuses on analyzing Spotify music trends to understand how these factors affect popularity and how music tastes have evolved over the years.

Importance of the Study

- Helps understand changing music trends and listener preferences
- Identifies key factors influencing song popularity
- Assists artists and producers in creating audience-preferred music
- Supports platforms like Spotify in improving recommendation systems
- Enables data-driven decision making in the music industry
- Provides insights useful for future music trend prediction

Objective of the Project

- To analyze Spotify music trends over the last decade
- To understand how audio features (energy, danceability, tempo, etc.) affect popularity
- To identify changes in listener preferences over time
- To explore relationships between genre, features, and song success
- To perform data cleaning, analysis, and visualization for meaningful insights
- To prepare data for future machine learning and prediction tasks

Scope of the Project

- Suggest Music directors on how people 's preferences over music change
- Focuses on analyzing Spotify music trends dataset over recent years
- Covers data cleaning, preprocessing, and exploratory data analysis (EDA)
- Studies relationships between audio features, genre, and popularity
- Identifies patterns and changes in listener preferences over time
- Applies feature engineering techniques to improve data quality
- Can be extended for machine learning models and recommendation systems in future

CHAPTER 2 -DATASET DESCRIPTION

Source of dataset: KAGGLE(SPOTIFY_MUSC_DATASETS_2023)

Dataset size (rows, columns)

```
df.shape
(2999, 19)
```

shape Displays the total number of rows and columns in the dataset.

Outcome: The output shows the dataset dimensions, which helps track changes after cleaning and outlier removal.

Description of features

```
df.describe()
```

	duration_ms	popularity	danceability	energy	key	loudness	mode	instrumentalness	tempo	stream_count	explicit
count	2999.000000	2999.000000	2999.000000	2999.000000	2999.000000	2999.000000	2999.000000	2999.000000	2999.000000	2.999000e+03	2999.000000
mean	251484.973324	48.304435	0.524001	0.504748	5.492497	-27.98994	0.510504	0.406277	129.180504	2.114361e+05	0.197733
std	95660.398382	15.019878	0.270222	0.278501	3.451154	15.73512	0.499973	0.232273	40.446089	1.642764e+06	0.398356
min	90096.000000	8.000000	0.050000	0.020000	0.000000	-54.99000	0.000000	0.000000	60.000000	1.000000e+03	0.000000
25%	166293.500000	38.000000	0.290000	0.270000	2.000000	-42.00000	0.000000	0.202000	94.640000	1.000000e+03	0.000000
50%	250553.000000	47.000000	0.520000	0.500000	6.000000	-27.66000	1.000000	0.404000	129.080000	2.000000e+03	0.000000
75%	334552.000000	57.000000	0.760000	0.740000	8.500000	-14.47000	1.000000	0.612000	164.370000	9.000000e+03	0.000000
max	419913.000000	100.000000	0.990000	0.990000	11.000000	-1.00000	1.000000	0.800000	199.980000	2.000000e+07	1.000000

`describe()` provides statistical information such as:

- Mean
- Standard deviation
- Minimum and maximum values
- Quartiles

for all numerical columns

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2999 entries, 0 to 2998
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
0   track_id              2999 non-null   object
1   track_name            2999 non-null   object
2   artist_name           2999 non-null   object
3   album_name            2998 non-null   object
4   release_date          2999 non-null   object
5   genre                 2999 non-null   object
6   duration_ms           2999 non-null   int64
7   popularity            2999 non-null   int64
8   danceability          2999 non-null   float64
9   energy               2999 non-null   float64
10  key                   2999 non-null   int64
11  loudness              2999 non-null   float64
12  mode                  2999 non-null   int64
13  instrumentalness       2999 non-null   float64
14  tempo                 2999 non-null   float64
15  stream_count          2999 non-null   int64
16  country               2999 non-null   object
17  explicit              2999 non-null   int64
18  label                 2999 non-null   object
dtypes: float64(5), int64(6), object(8)
memory usage: 445.3+ KB
```

info() shows:

- Total number of rows
- Column names
- Data types
- Number of non-null values

Initial data issues

Missing values:

spotify_data.isnull().sum()

...	0
track_id	0
track_name	0
artist_name	0
album_name	1
release_date	0
genre	151
duration_ms	267
popularity	282
danceability	285
energy	281
key	301
loudness	273
mode	307
instrumentalness	313
tempo	289
...	...

Outlier detection:

```
outlier_cols = df.select_dtypes(include=np.number).columns
z_scores = pd.DataFrame(
    np.abs(stats.zscore(df[outlier_cols])),
    columns=outlier_cols,
    index=df.index
)
df['outlier_flag'] = (z_scores > 3).any(axis=1)

df['outlier_columns'] = z_scores.apply(
    lambda row: list(outlier_cols[row > 3]),
    axis=1
)

df['outlier_flag'].value_counts()
```

...	count
outlier_flag	
False	2826
True	173

dtype: int64

CHAPTER 3-DATA PREPROCESSING AND CLEANING

Uploading dataset:

```
[ ] from google.colab import files

    uploaded = files.upload()

    for fn in uploaded.keys():
        print(f'User uploaded file "{fn}" with length {len(uploaded[fn])} bytes')
```

... Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser

Saving spotify.csv to spotify (1).csv
User uploaded file "spotify (1).csv" with length 459827 bytes

```
[ ] import pandas as pd
    import numpy as np
    import matplotlib.pyplot as plt
    from scipy import stats
```

```
[ ] df = pd.read_csv("spotify.csv")
```

Handling Missing values:

```
num_cols = df.select_dtypes(include=np.number).columns
cat_cols = df.select_dtypes(exclude=np.number).columns

df[num_cols] = df[num_cols].fillna(df[num_cols].mean())

for col in cat_cols:
    df[col].fillna(df[col].mode()[0], inplace=True)
```

Justification for Using Mean Imputation

- Mean imputation replaces missing values with the average of the column, helping maintain the overall data distribution
- It is suitable for numerical features like energy, tempo, and popularity
- It ensures that no data is lost, unlike removing rows with missing values
- It keeps the dataset balanced and consistent for analysis
- It is simple, efficient, and works well when the data is not highly skewed

Outlier Detection

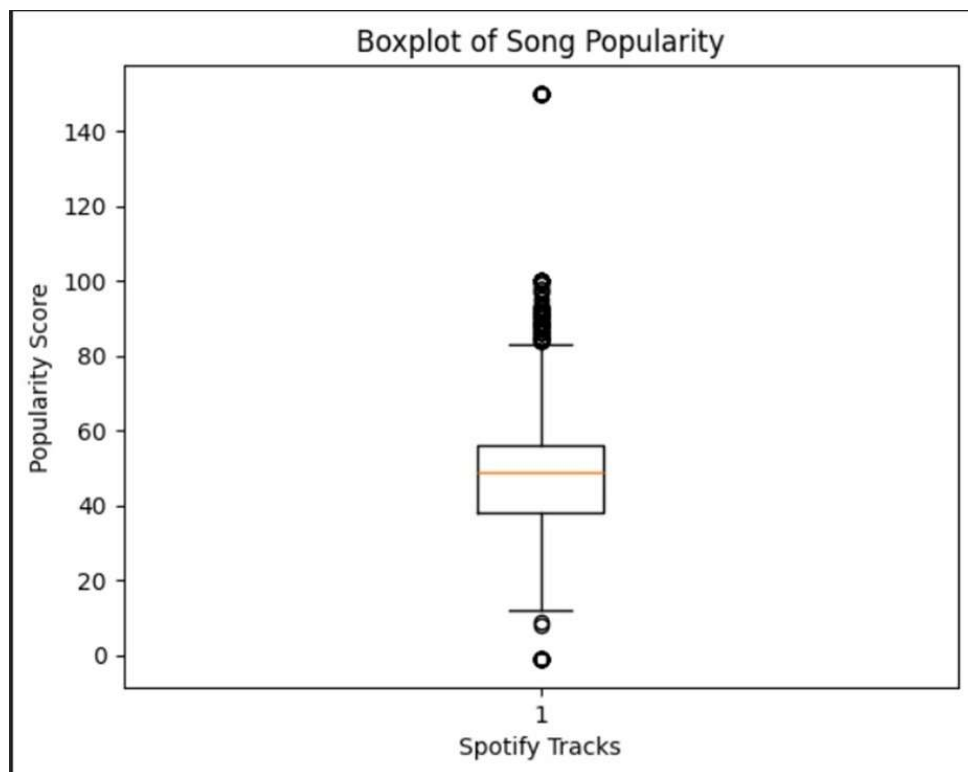
```
outlier_cols = df.select_dtypes(include=np.number).columns
z_scores = pd.DataFrame(
    np.abs(stats.zscore(df[outlier_cols])),
    columns=outlier_cols,
    index=df.index
)
df['outlier_flag'] = (z_scores > 3).any(axis=1)

df['outlier_columns'] = z_scores.apply(
    lambda row: list(outlier_cols[row > 3]),
    axis=1
)

df['outlier_flag'].value_counts()
```

```
...          count
outlier_flag
False        2826
True         173

dtype: int64
```

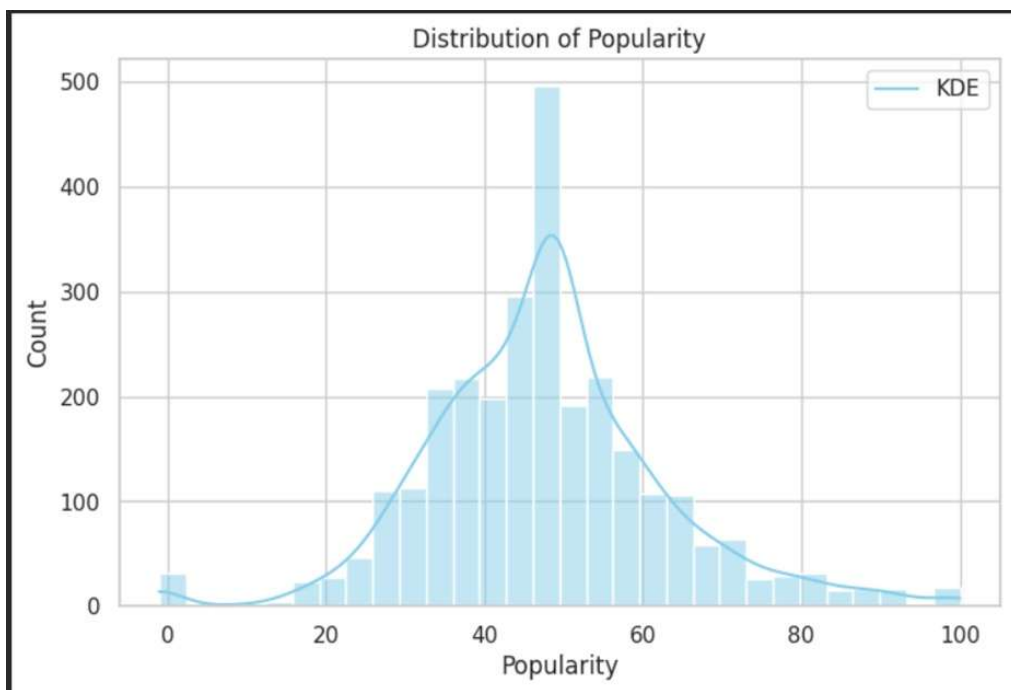


Report Summary:

- Missing values handled effectively using imputation techniques
- Outliers detected and treated using boxplots and Z-score methods
- Duplicate records removed and data types corrected
- A clean, structured Spotify dataset ready for exploratory data analysis and visualization

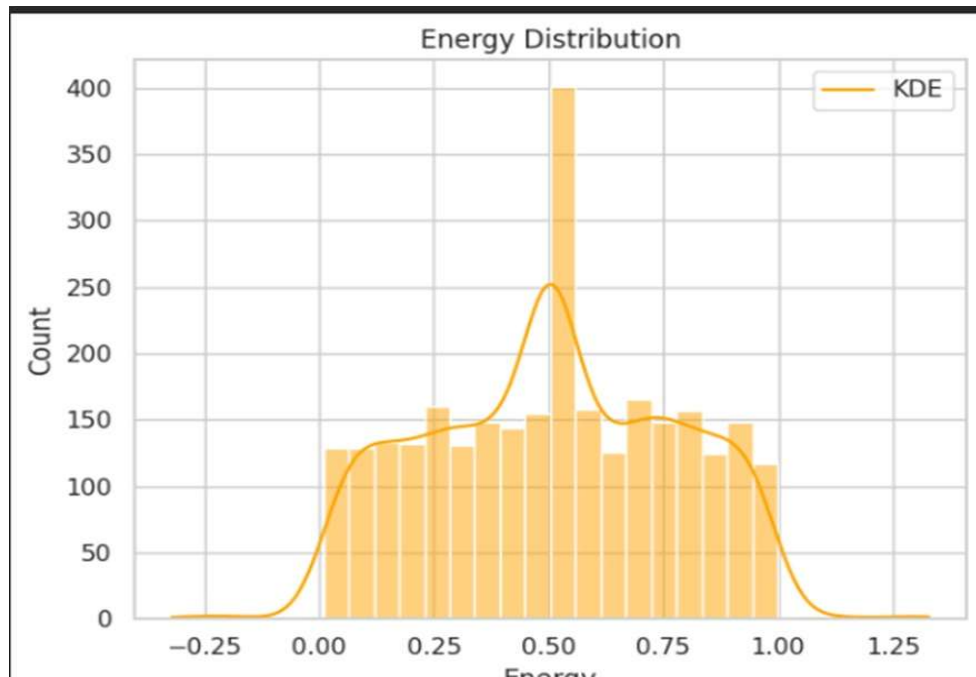
CHAPTER 4- EXPLORATORY DATA ANALYSIS(EDA)

Univariate analysis (distribution plots)



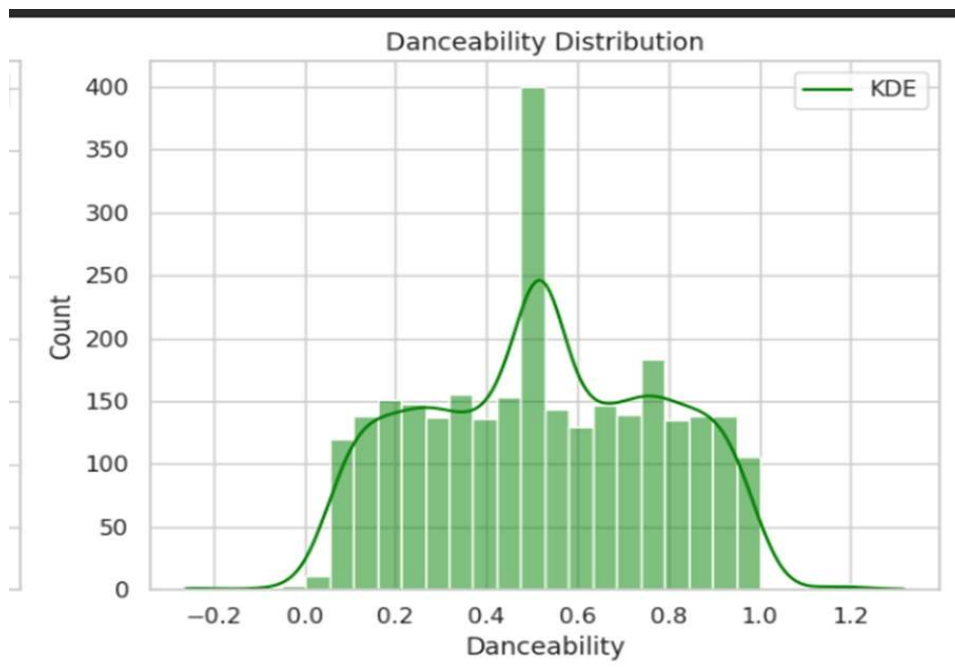
Popularity distribution

- The distribution of popularity appears approximately bell-shaped (near normal distribution).
- Most songs have popularity values concentrated in the mid-range (around 40–60).
- Very low (0–10) and very high (90–100) popularity values are rare.
- Slight right skew may be present, indicating a few highly popular songs.



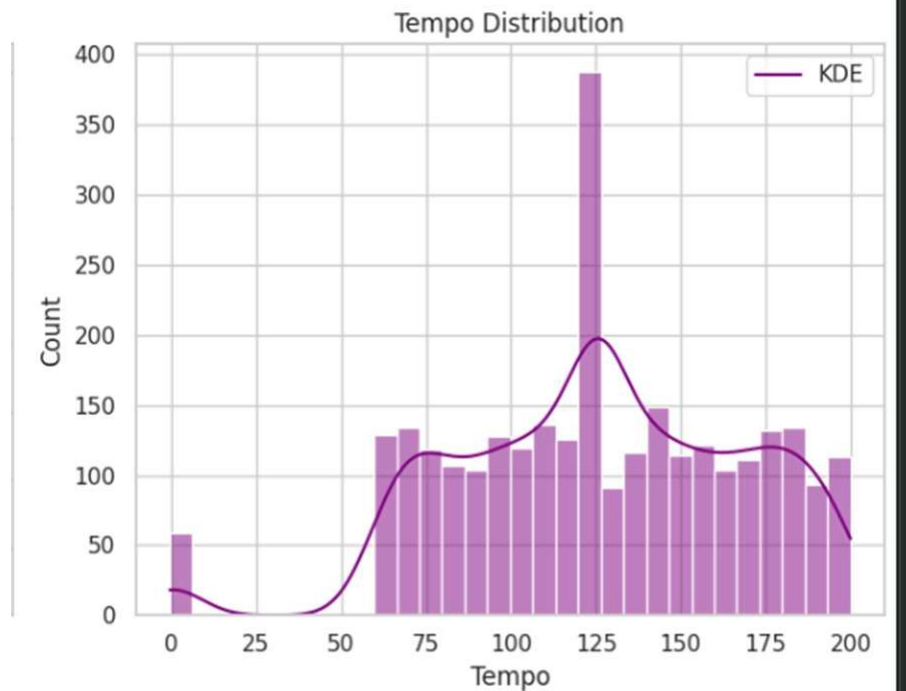
Energy

- Values are likely concentrated in the mid to high range (0.5–0.8).
- Indicates most songs are moderately to highly energetic.
- Few songs have extremely low energy.



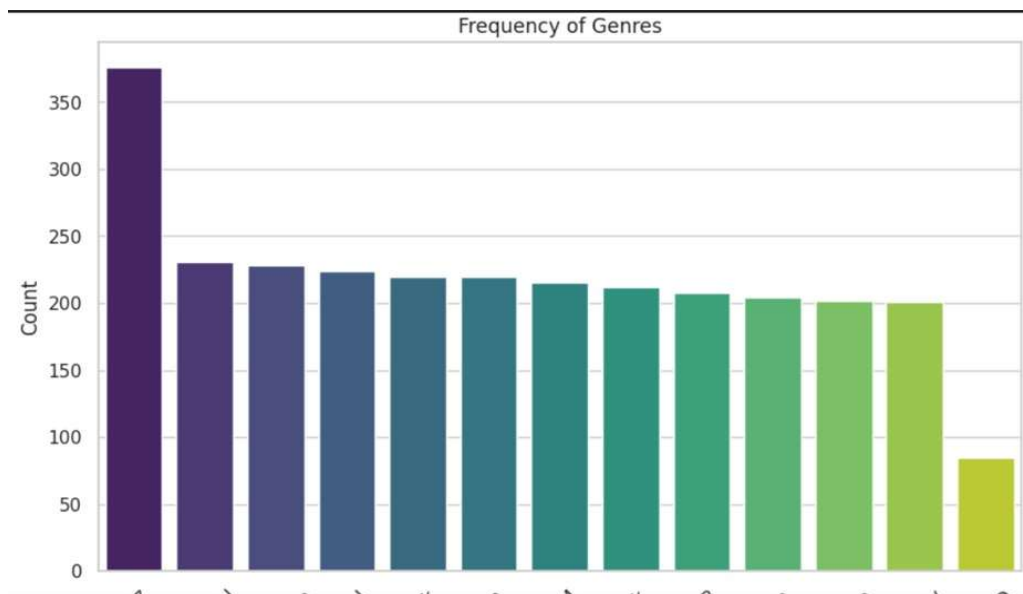
Danceability

- Distribution likely centered around 0.5–0.8.
- Shows that many songs are reasonably danceable.
- Few songs are extremely low in danceability.



Tempo

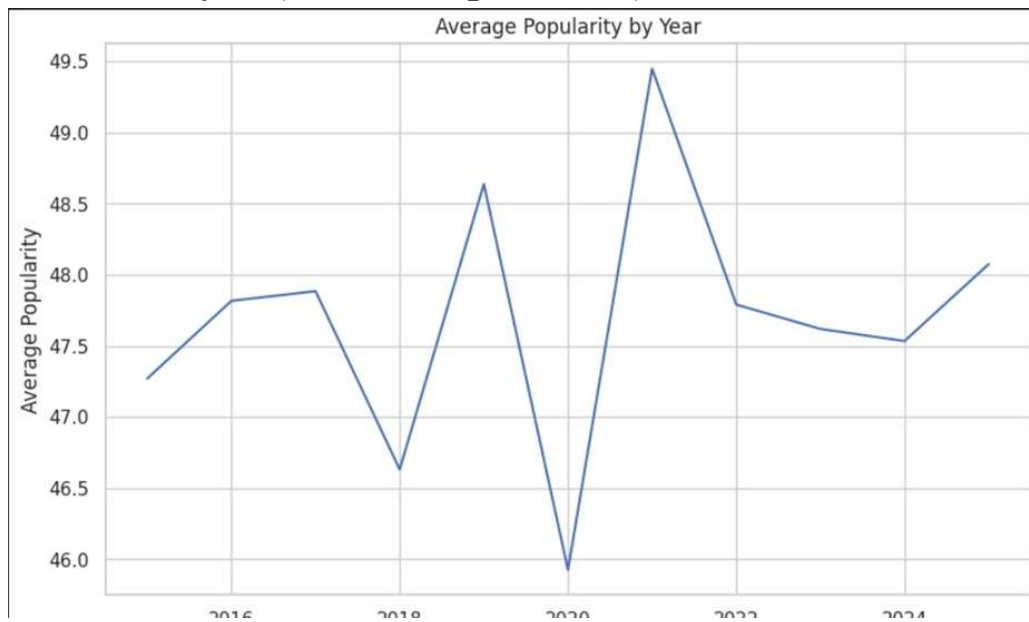
- Tempo shows wider spread compared to energy/danceability.
- Likely centered around 100–140 BPM.
- May show slight right skew due to high-tempo songs



Genre Frequency

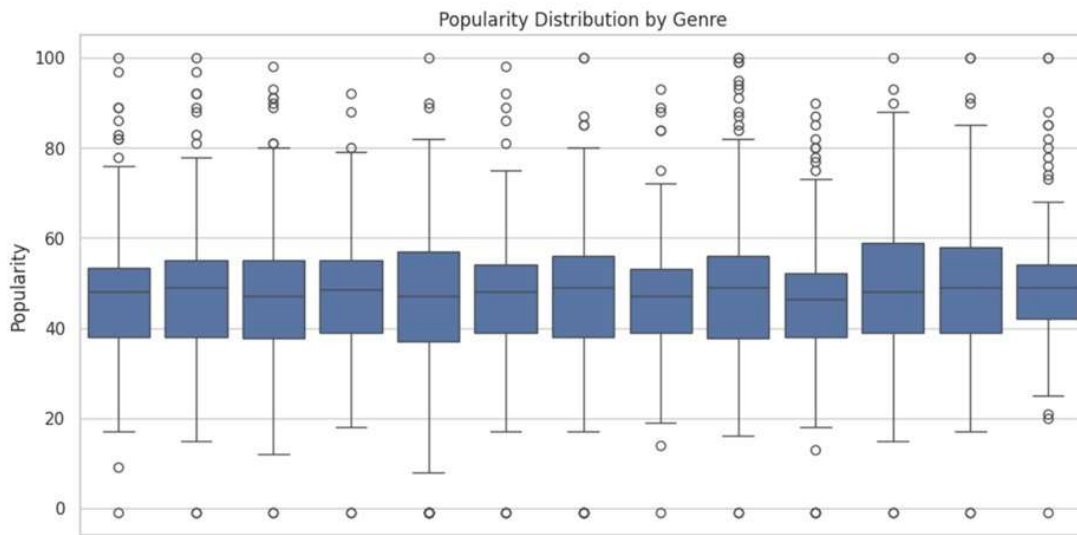
- Shows how many songs belong to each genre.
- Some genres have significantly more songs than others.
- Indicates dataset imbalance in genre distribution

Bivariate analysis (relationships, trends)



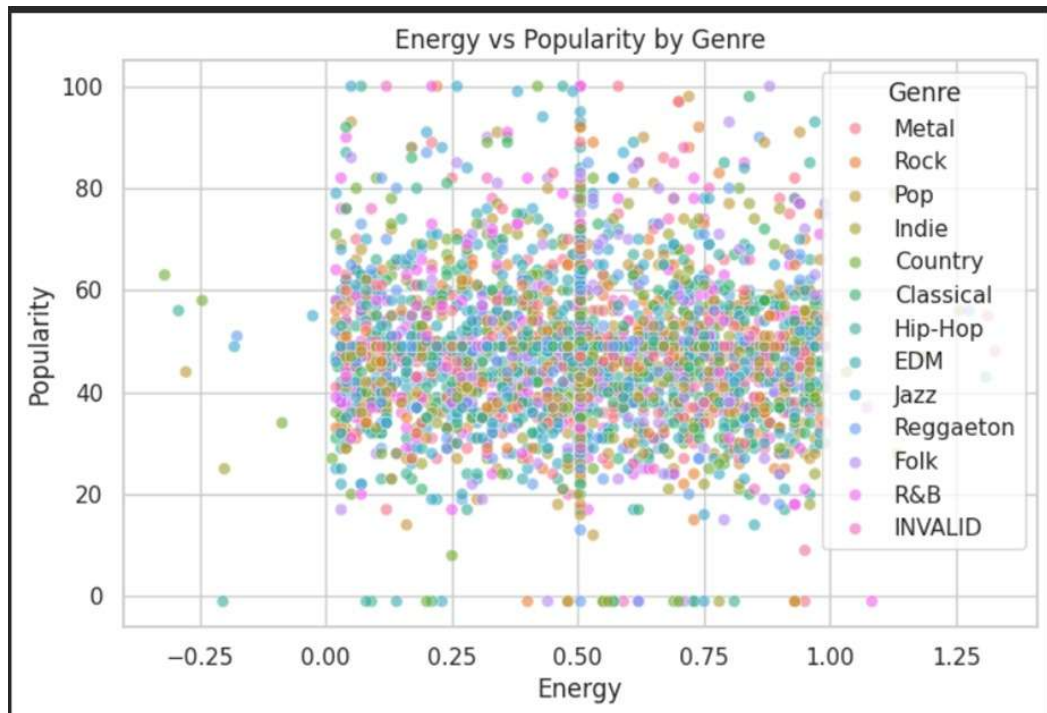
Popularity distribution by year

- Shows trend of average popularity over time.
- If upward trend → newer songs are more popular.
- If fluctuating → popularity varies across years.
- No error bars means it shows only mean trend



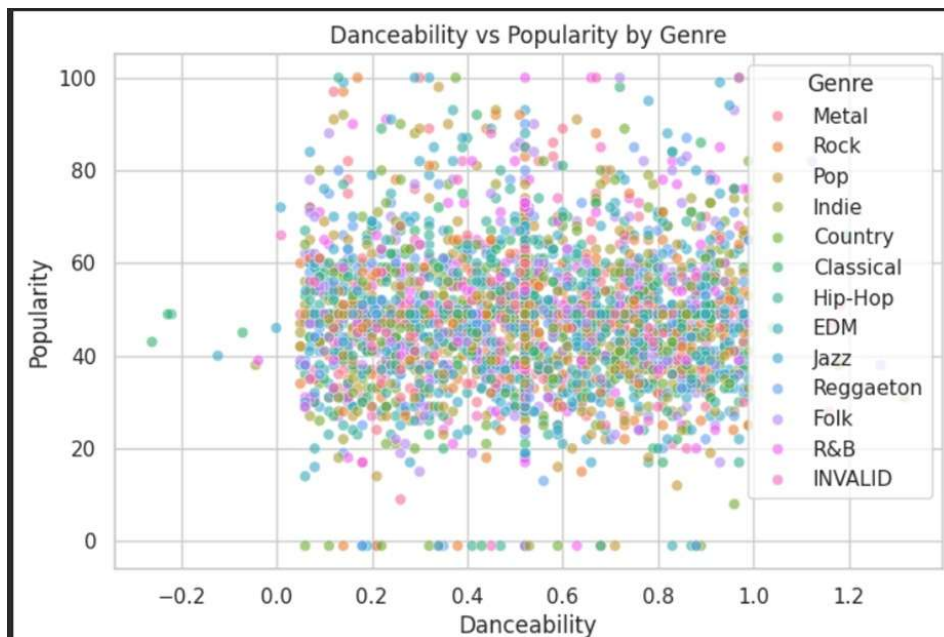
Popularity distribution by genre

- Shows median, quartiles, and outliers of popularity for each genre.
- Some genres may have:
- Higher median popularity
- Wider spread (high variability)
- More outliers (hit songs)



Energy Vs Popularity by genre

- Shows relationship between energy and popularity.
- If points trend upward → higher energy songs are more popular.
- If scattered randomly → weak correlation.
- Coloring by genre shows which genres dominate certain regions.



Danceability vs Popularity by genre

- Shows whether danceable songs are more popular.
- If upward pattern → positive correlation.
- Likely weak-to-moderate positive relationship.

CHAPTER 5 – DATA TRANSFORMATION AND FEATURE ENGINEERING

Feature Classification:

```
num_cols = df.select_dtypes(include=['int64', 'float64']).columns
cat_cols = df.select_dtypes(include=['object', 'bool']).columns

print("Numerical Columns:", num_cols)
print("Categorical Columns:", cat_cols)

Numerical Columns: Index(['duration_ms', 'popularity', 'danceability', 'energy', 'key',
                          'loudness', 'mode', 'instrumentalness', 'tempo', 'stream_count'],
                        dtype='object')
Categorical Columns: Index(['track_id', 'track_name', 'artist_name', 'album_name', 'release_date',
                          'genre', 'country', 'explicit', 'label', 'outlier_flag'],
                        dtype='object')
```

Applied Standard Scaling:

- Used StandardScaler and
- Scaled all numerical features

Purpose:

- Normalize data
- Improve ML model performance

```
scaler = StandardScaler()
df[num_cols] = scaler.fit_transform(df[num_cols])
```

Encoding techniques (Label / One-hot)

Applied Label Encoding:

- Converted categorical data into numerical form

Used: LabelEncoder

```
le = LabelEncoder()

for col in cat_cols:
    df[col] = le.fit_transform(df[col].astype(str))
```

Skewness Handling

```
skew_values = df[num_cols].apply(lambda x: skew(x.dropna()))  
print(skew_values.sort_values(ascending=False))
```

```
stream_count      2.510866  
popularity         0.288653  
instrumentalness  -0.000346  
loudness          -0.008313  
danceability      -0.010997  
key               -0.016111  
energy            -0.021096  
duration_ms      -0.236209  
mode              -0.271264  
tempo             -0.372845  
dtype: float64
```

Skewness Detection-Calculated skewness using:

`scipy.stats.skew`

Sorted skew values to identify highly skewed features

Applied Power Transformation (Yeo-Johnson):

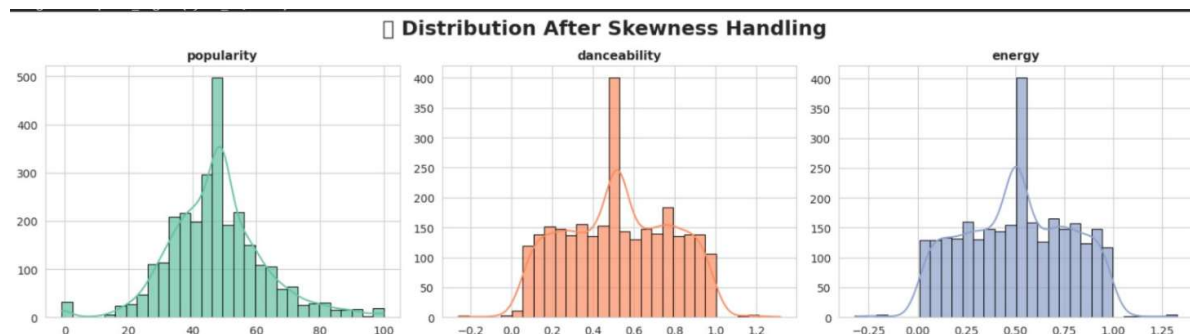
- Specifically on tempo feature

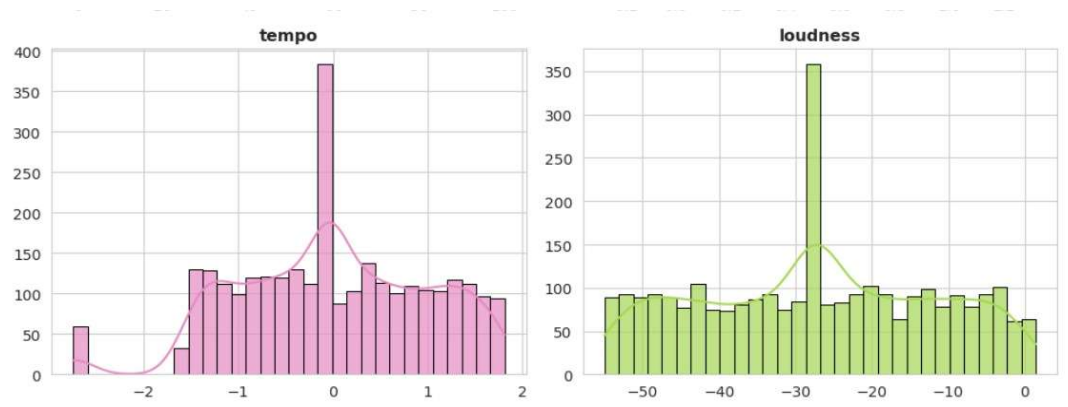
Purpose:

- Reduce skewness
- Make data more normally distributed

```
pt = PowerTransformer(method='yeo-johnson')  
df['tempo'] = pt.fit_transform(df[['tempo']])
```

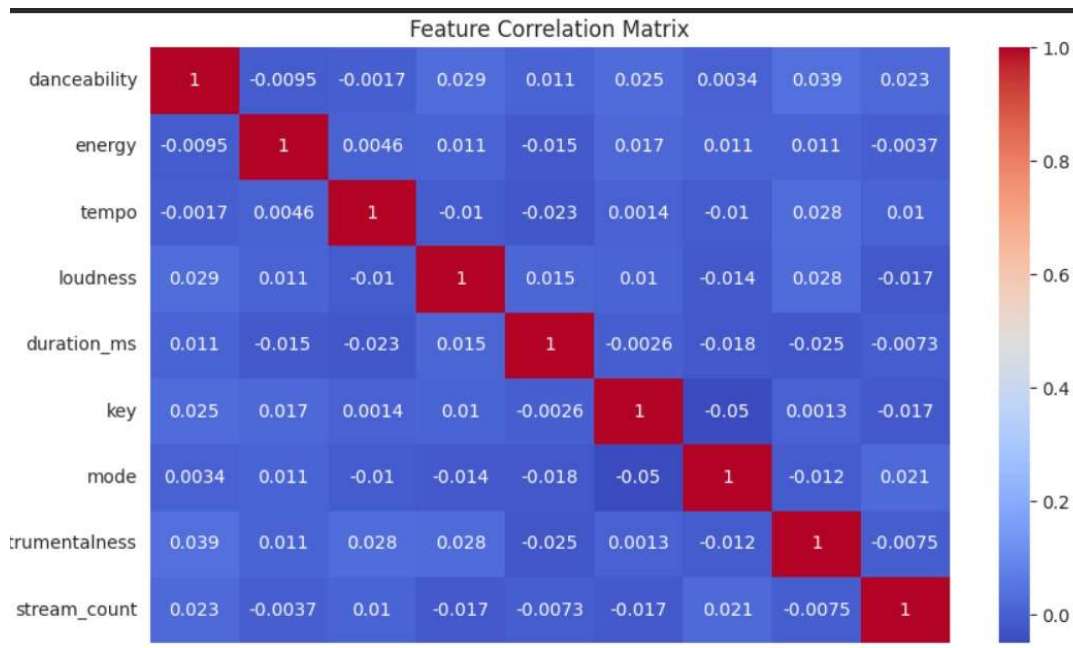
Visualization(after handling skewness)





CHAPTER 6 – ADVANCED ANALYSIS

Correlation between audio features



Feature correlation Matrix:

- Shows relationships between different music features.
- No clear upward or downward patterns → correlations are very weak.
- Most values are close to zero → little to no relationship.
- Stream count vs features → likely no strong impact on popularity.

Feature Selection:

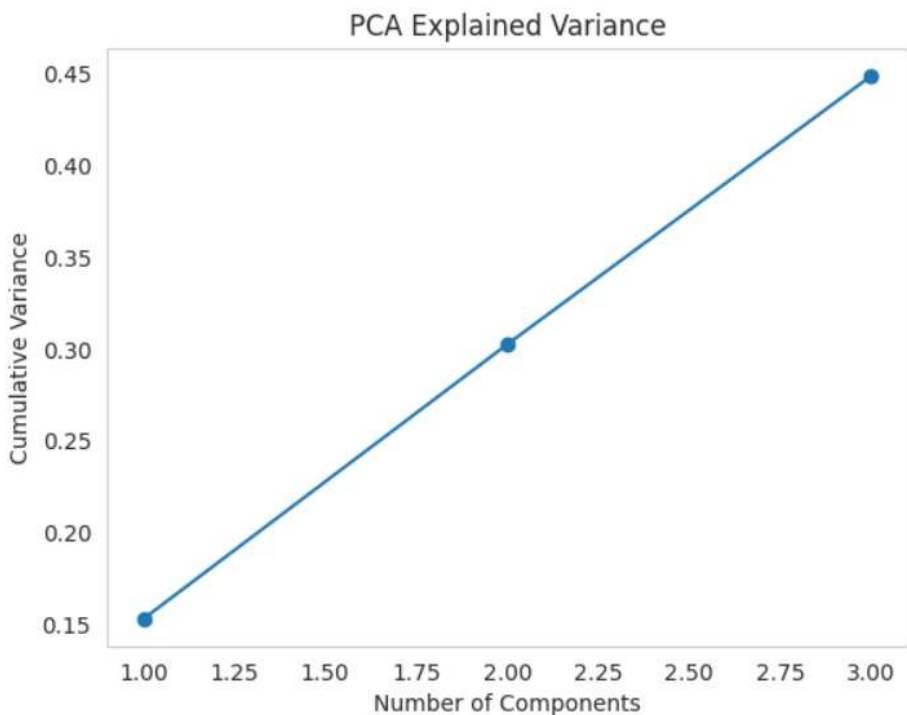
```
features_before = ['danceability', 'energy', 'tempo', 'loudness',
                  'duration_ms', 'instrumentalness', 'stream_count']
```

Principal Component Analysis (PCA)

Before Reduction Shape: (2826, 7)
Features Before: ['danceability', 'energy', 'tempo', 'loudness', 'duration_ms', 'instrumentalness', 'stream_count']
After Reduction Shape: (2826, 3)
Number of components: 3

Features After Reduction:
['PC1', 'PC2', 'PC3']

	PC1	PC2	PC3
0	-0.413865	0.760375	-0.285754
1	-0.992419	0.537966	-0.389508
2	0.047828	-0.723066	-0.146436
3	0.623236	-0.254862	0.465541
4	1.240152	1.351691	0.186877



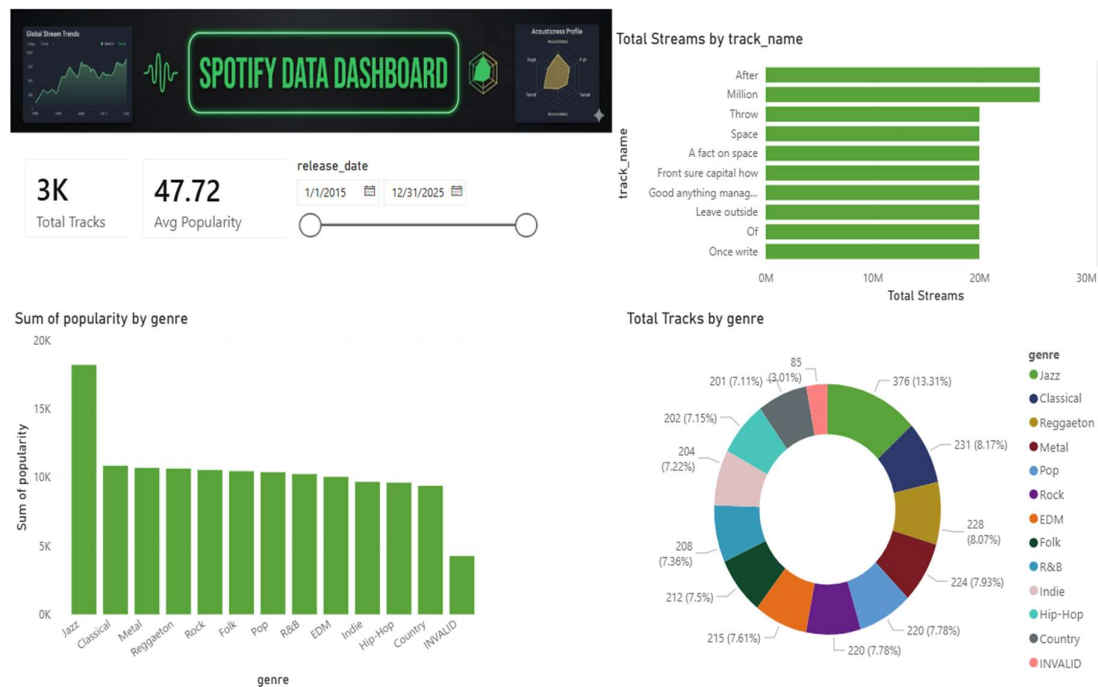
PCA Model MSE: 269.63646692599184
PCA Model R2 Score: -0.15703839208394932

CHAPTER 7-VISUALIZATION AND DASHBOARD

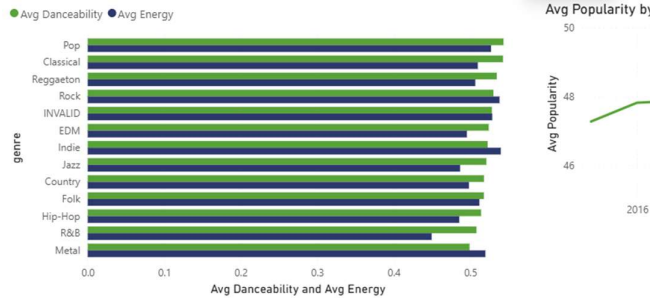
Tools used: Power BI

- Used Microsoft Power BI to create interactive and visually appealing dashboards
- Helps in real-time data visualization and easy understanding of trends like popularity, genre, and audio features
- Enables quick insights and better decision-making through dynamic charts, filters, and reports

SPOTIFY DATA DASHBOARD

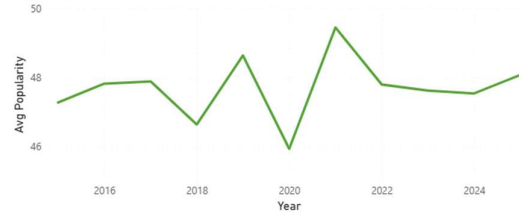


Avg Danceability and Avg Energy by genre

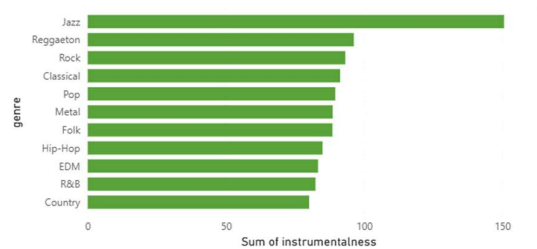


Rename file

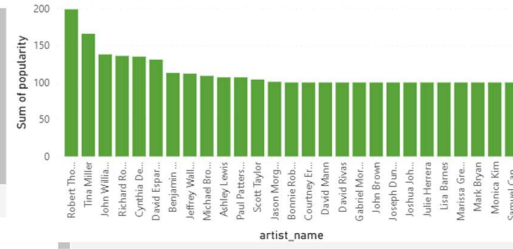
Avg Popularity by Year



Sum of instrumentality by genre



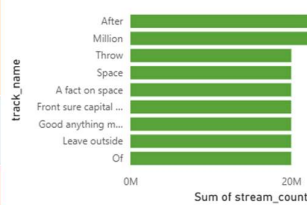
Sum of popularity by artist_name



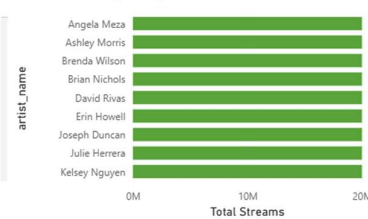
Total Streams by genre



Sum of stream_count by track_name

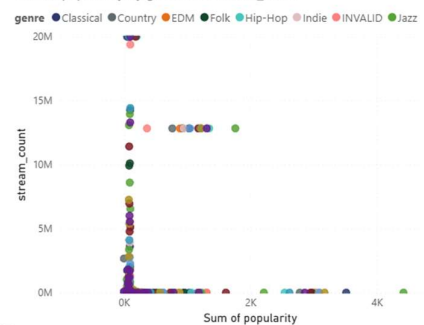


Total Streams by artist_name



artist_name	track_name	Sum of stream_count	Sum of popularity	outlier_flag
Samuel Cantrell	Although him	1296000	100.00	False
David Mann	Draw	5122000	100.00	False
Bonnie Roberson	Finally easy exist	6555000	100.00	False
Marissa Green	Front sure capital how	20000000	100.00	False
Julie Herrera	Good anything manager think	20000000	100.00	False
Mark Bryan	Ground once design walk simply	2013000	100.00	False
David Rivas	Of	20000000	100.00	False
Lisa Barnes	Outside former official	19380000	100.00	False
Gabriel Morton	Pattern friend	14422000	100.00	False
Courtney Erickson	Size writer	12831004	100.00	False
Joseph Duncan	Space	20000000	100.00	False
Monica Kim	Upon	0	100.00	False
Joshua Walton	Cost group	199000	99.00	False
Tina Gregory	Whose	12831004	99.00	False
Erica Snow	Perhaps recent	2256000	98.00	False
Catherine Smith	Security economic	14254000	98.00	False
Stephen Ho	Animal draw although	13296000	97.00	False
Angela Meza	Whose section	20000000	97.00	False
Total		4324289180	134,847.91	

Sum of popularity by genre and stream_count



1. Overall Summary (Top Section)

- Total Tracks: ~3K
- Average Popularity: ~47.7
 - Shows the dataset size and overall popularity level (moderate range)
- Date Filter (2015–2025)
 - Allows analysis of music trends across years

2. Total Streams by Track

- Displays top streamed songs
- Songs like “*After*”, “*Million*” have highest streams (~20M+)
Indicates most popular tracks based on audience listening

3. Popularity by Genre

- Jazz has highest total popularity
- Followed by Classical, Metal, Reggaeton, Rock
Shows which genres are most preferred

4. Total Tracks by Genre (Donut Chart)

- Jazz has highest share (~13%)
- Other genres are evenly distributed
Dataset has slight imbalance but good coverage

5. Danceability & Energy by Genre

- Most genres show moderate to high values (~0.5)
- Pop, Rock, Reggaeton → slightly higher
- Modern music is generally energetic and danceable

6. Popularity Trend Over Years

- Slight fluctuations with peak around 2021
- No steady increase/decrease
Music popularity is dynamic over time

7. Instrumentalness by Genre

- Jazz has highest instrumentalness
- Others are lower
Jazz focuses more on instrumental music

8. Popularity by Artist

- Few artists dominate with higher popularity
Shows uneven artist success distribution

9. Streams by Genre (Treemap)

- Jazz occupies largest area
Confirms Jazz dominates in streams

10. Scatter Plot (Popularity vs Streams)

- Points are scattered → weak correlation
High streams ≠ always high popularity

CHAPTER 8 – INSIGHTS AND STORYTELLING

Key Findings from Analysis

- Most songs have moderate popularity (40–60)
- Songs are generally high in energy and danceability
- Jazz and few other genres dominate in popularity and streams
- No single feature strongly determines popularity
- Data shows weak correlation between streams and popularity
- Dataset required cleaning and feature engineering for better accuracy

Patterns and Trends Identified

- Music preferences are dynamic and change over years
- Popular songs tend to be energetic and danceable
- Genre imbalance exists (some genres more frequent)
- Instrumental music is higher in Jazz
- Popularity shows fluctuation, not steady growth
- Relationships between features are mostly weak to moderate

Business / Social Insights

- Audience prefers engaging and upbeat music
- Music industry is trend-driven and continuously evolving
- Platforms like Spotify can improve recommendations using audio features
- Artists can focus on popular genres and listener behaviour
- Listener choices reflect cultural and social influences

Decision Making based on results:

Highly preferred genre:

- JAZZ
- Other genres remains less in popularity

Popular Artist:

- Robert Thompson: Rock(199)
- Richard Roduriguez : Jazz (136)

High popular year: 2021

CHAPTER 9 – CONCLUSIONS

Summary of the Project

This project focuses on analyzing Spotify music trends using a real-world dataset to understand how music preferences have evolved over time. It involves data preprocessing, handling missing values using mean imputation, outlier detection, and exploratory data analysis (EDA). The study examines relationships between audio features like energy, danceability, tempo, and popularity. Feature engineering techniques were also applied to improve data quality. The project successfully identifies patterns, trends, and key factors influencing music popularity.

Future Scope

- Apply machine learning models to predict song popularity
- Build a music recommendation system
- Use deep learning techniques for advanced pattern detection
- Include real-time streaming data for dynamic analysis
- Expand dataset with more features like lyrics, mood, and user behavior
- Develop interactive dashboards for better visualization.

Outcomes Achieved

- Successfully cleaned and preprocessed the dataset
- Handled missing values and outliers effectively
- Identified music trends and listener preferences
- Found relationships between audio features and popularity
- Performed univariate, bivariate, and multivariate analysis
- Improved dataset quality using feature engineering techniques
- Generated meaningful insights for data-driven decisions

Limitations

- Dataset may have genre imbalance affecting analysis
- Mean imputation may not fully represent true missing values
- Analysis is limited to available features only
- No implementation of predictive models
- Results depend on historical data, may not reflect future trends
- External factors (social media, marketing) not considered

CHAPTER 10 – REFERENCES

Dataset Sources

Spotify Music Trends Dataset (Kaggle)

Dataset link:

<https://drive.google.com/file/d/1Tg8PsIE0QLwRdZzDV8T1nD1aMzpvKsk/view?usp=drivesdk>

Contains features like genre, popularity, energy, danceability, tempo, loudness, year

Tools and Libraries Used

Tools:

Google Colab / Jupyter Notebook

Python

GitHub / OneDrive (for storage & sharing)

Libraries:

pandas – Data handling

numpy – Numerical operations

matplotlib – Basic visualization

seaborn – Advanced visualization

scipy – Statistical methods (Z-score)

sklearn (scikit-learn) – Scaling, encoding, transformations

Research Papers / Websites Referred

- Kaggle – Spotify Dataset & discussions
- Official pandas documentation
- Official NumPy documentation
- Scikit-learn documentation
- Matplotlib & Seaborn documentation
- Google Colab documentation
- Online tutorials and articles on EDA and data preprocessing

Appendix

GitHub link:

https://github.com/JANASREE-P/24ADI004_DSV_TEAM4.git